A Cross-Description Test of Sparse Feature Correspondence Between Language Models and fMRI Responses

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Paper under double-blind review

Abstract

Language-model features predict fMRI responses to spoken language, but shared prediction does not show that the two systems organize those features the same way. We test the stronger claim directly. Sparse-autoencoder (SAE) coordinates are selected from one spoken description of an animation, used to fit separate ridge mappings to cortical BOLD and to the language model's own hidden state, and then transferred, unchanged, to a second description of the same animation. For each coordinate we measure how much held-out prediction each mapping loses when that coordinate is removed. If BOLD and the model rely on the sparse basis in a corresponding way, these two loss profiles should agree, and should agree more than they do when the model target is shifted to the wrong narrative time. Across six model–SAE systems the basis predicts both targets, though BOLD only weakly ($r = .07$–$.09$ versus .30–.48 for the model state). The loss profiles do not agree: at $K = 256$ features the mean rank correlation across all twelve model–direction conditions is $\rho = -.056$, and 177 of 199 fully retuned temporal-null means are at least as large ($p = .890$). Changing the bank size, pruning correlated features, ablating correlated features as groups, changing token pooling, and averaging participants all leave that conclusion unchanged; the one favorable cell does not replicate in the reverse direction or across model families. A semi-synthetic calibration on one model family quantifies what this failure can mean: the full pipeline detects an injected correspondence of $\rho = .40$ with only .22–.35 probability and reaches the prespecified 80% power only near $\rho = .80$, while its false-positive rate is exactly the nominal .05. The null result therefore bounds strong correspondence for that family but cannot distinguish moderate correspondence from absence. Prediction alone does not establish feature correspondence, and the present design's resolution, now measured, is what limits the stronger negative claim.

1 INTRODUCTION

Autoregressive language models are trained to predict the next token, yet their internal activations predict human brain responses during language comprehension, and models that predict text better tend to predict the brain better (Schrimpf et al., 2021; Kumar et al., 2024). This result has been read as evidence that the two systems share representational structure. But an encoding score measures only whether a model's activations carry information that a linear map can exploit. Word rate, acoustic energy, position in the narrative, and the slow temporal autocorrelation of any story-driven signal can all support prediction (Hadidi et al., 2026). A single encoding score therefore cannot tell us whether the two systems depend on *corresponding* internal features. That is a separate question, and it needs a separate measurement.

Sparse autoencoders (SAEs) make the question operational. They decompose a dense model state into sparse coordinates that can be traced to the text that activates them (Cunningham et al., 2023; Lieberum et al., 2024). The same named coordinate can be supplied to two prediction problems, one targeting BOLD and one targeting the model's own hidden state, and then removed from each. If the coordinate matters to both fitted mappings, and if that pattern repeats across many coordinates, the two targets share an ordering of predictive reliance. This turns a broad alignment claim into a concrete question about held-out performance.

Two features of the design keep the answer from being trivial. First, both mappings see the same story at the same times, so a positive ordering can arise from shared temporal structure alone; the test must show that the observed ordering beats one obtained when the model target is deliberately misaligned in time and the whole pipeline is retuned. Second, SAE coordinates are correlated and can substitute for one another in a ridge fit, so

single-coordinate ablation can understate reliance; the test must also be run on correlated groups. We add a third safeguard: features are chosen from one description of a stimulus and evaluated, without refitting, on a second description of the same events, so the comparison is a transfer test rather than an in-sample fit.

We run this test on two spoken descriptions of one short animation from the Narratives dataset (Nastase et al., 2021), heard by the same 59 participants, with six model–SAE systems spanning three model families and 1B to 12B parameters. The paper answers four questions in turn. Does a source-selected sparse basis predict both targets on the second description? Do the two mappings lose performance in the same feature order? Does that agreement survive nulls that break temporal alignment or exact feature identity? And do redundancy, bank size, token pooling, or BOLD noise change the answer?

The answers are yes, no, no, and no. The basis transfers. The reliance orderings do not agree: their mean rank correlation is slightly negative, no better than the fully retuned temporal null, and no analysis choice we tested makes it positive. The paper's contribution is therefore a test and a bound rather than a discovery: with source-only selection, cross-description transfer, and structure-preserving nulls in place, held-out prediction does not carry over into a transferable ordering of feature importance. Whether that reflects a true absence or a limit of measurement resolution is the one question the present design leaves open, and we say precisely what would close it.

2 Related work

Encoding models and what they measure. Encoding models relate stimulus features to brain activity during naturalistic perception (Huth et al., 2012; Wehbe et al., 2014). With language-model predictors, the resulting correlation is often called neural predictivity or brain alignment (Schrimpf et al., 2021; Kumar et al., 2024; Chen and Sivakumar, 2026). The label is broad. Hadidi et al. (2026) show that apparently strong alignment can arise from non-robust evaluation and overlooked confounds. We follow their separation of prediction from interpretation, but ask a different question: not whether an alignment score is inflated, but whether a successful score implies corresponding feature organization at all.

Sparse features and correspondence. SAEs decompose activations into sparse features with a learned decoder (Cunningham et al., 2023; Lieberum et al., 2024; He et al., 2024). The features are useful for inspection but are not canonical: their identities depend on model, layer, training data, width, and seed (Leask et al., 2025). Brain–SAE studies in language and visual settings show that sparse features can support interpretable encoding and cortical mapping (Lepori et al., 2026; Guo et al., 2026; Mao et al., 2025). Our question is narrower. We ask whether BOLD and the model's own state depend on the same sparse coordinates in the same order, after stimulus, feature identities, and model fitting have been held fixed across the two targets.

Why held-out ablation rather than coefficients. A fitted linear model can distribute weight among correlated predictors in many ways, so a large coefficient need not identify an important coordinate, and coefficient magnitude scales with predictor units. Conditional held-out ablation asks the operational question directly: after the model has been fit, how much predictive performance is lost when a coordinate is unavailable? Grouped ablation asks the same of correlated families. Neither quantity identifies a biological mechanism, but together they show whether an apparent correspondence depends on an arbitrary division of redundant predictors.

3 Method

3.1 Stimuli, participants, and models

We use SHAPESPHYSICAL and SHAPESSOCIAL from the Narratives dataset (Nastase et al., 2021; OpenNeuro, 2021): two spoken descriptions, differing in words and timing, of the same short animation. Both were heard by the same 59 eligible participant-runs, sampled every 1.5 s, giving 303 and 305 time bins (TRs). The BOLD

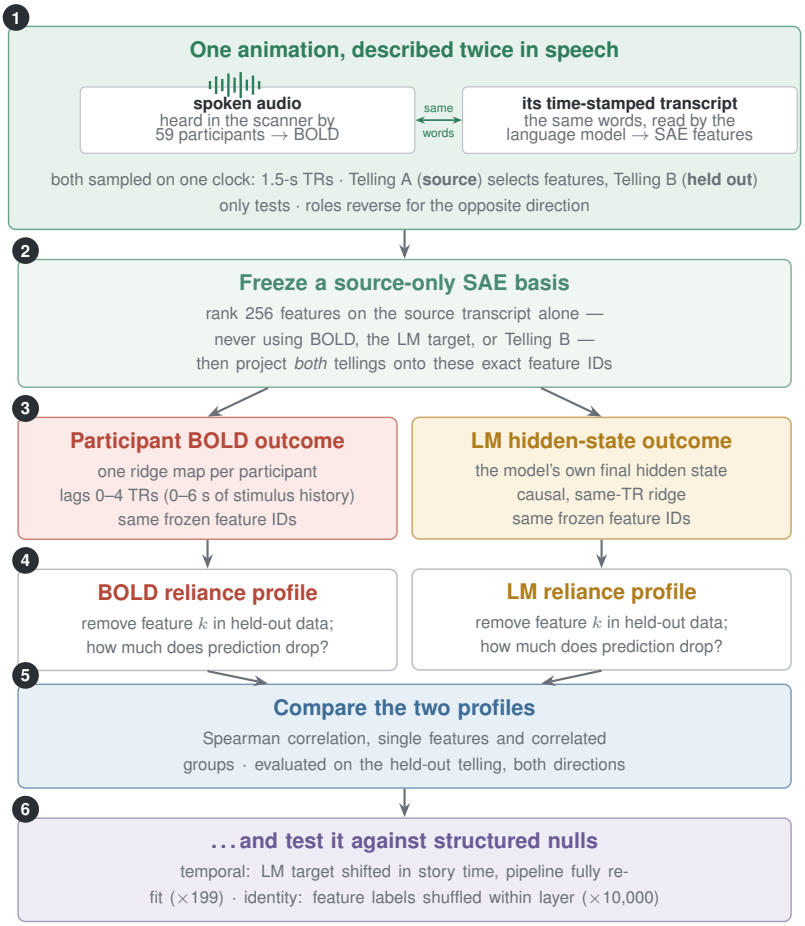


Figure 1: The experiment in one pass. The stimulus exists in two forms: spoken audio, heard in the scanner by 59 participants, and its time-stamped transcript, read by the language model — both sampled on the same 1.5-s TR grid (step 1). A 256-feature SAE basis is frozen using the source transcript alone (step 2) and feeds two prediction problems on the same coordinates (step 3). Removing each feature in held-out data yields one reliance profile per outcome (step 4); the profiles are compared on the held-out telling in both directions (step 5), against nulls that break target timing or exact feature identity (step 6).

target is the cleaned, run-standardized response in 200 cortical parcels (Schaefer et al., 2018). Cohort construction, preprocessing, and censoring are in Appendices A and B.

Six model–SAE systems are evaluated: Gemma 2 2B and 9B (Gemma Team, 2024; Lieberum et al., 2024), Llama 3.1 8B (Grattafiori et al., 2024; He et al., 2024), and Gemma 3 1B, 4B, and 12B (Google DeepMind, 2025a; Gemma Team, 2025), each with its official residual-stream SAE release (Lieberum et al., 2024; He et al., 2024; Google DeepMind, 2025b) at six depth-spaced layers (Appendix D). Token-level SAE activations are mapped to timed words and averaged within each TR. The model target at TR t is the mean final hidden state over the tokens assigned to that TR, computed in causal windows of at most 1,024 tokens.

This model target is deliberately model-internal. It asks which sampled SAE coordinates linearly predict the model's final state on the same time grid, and how a ridge surrogate for that prediction depends on them. It does not intervene on the model or measure contribution to next-token loss or behavior. A brain profile could in principle correspond to a behaviorally grounded target while failing to correspond to this one; we return to this in Section 5.

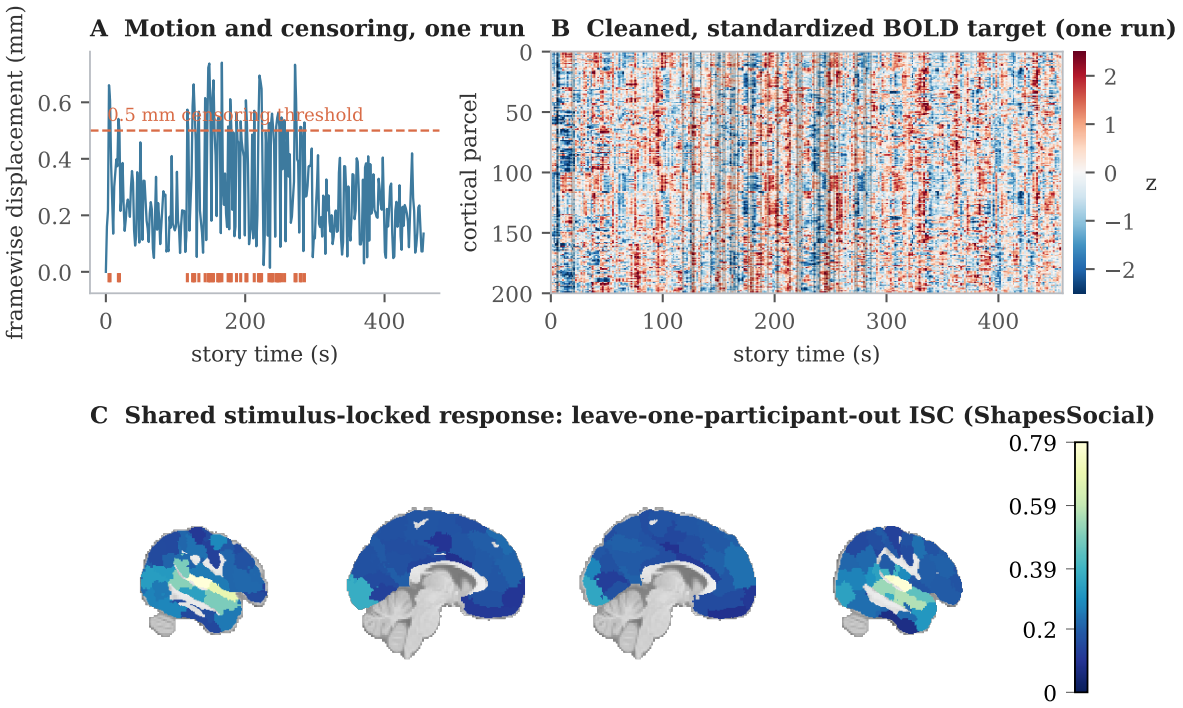


Figure 2: The measurement behind the BOLD target. **A:** Framewise displacement for the highest-motion retained SHAPESSOCIAL run; orange ticks mark TRs flagged by the motion, DVARS, or non-steady-state criteria and excluded as response rows. **B:** The same run's cleaned, run-standardized 200-parcel target; shaded columns are censored TRs. **C:** Median leave-one-participant-out inter-subject correlation per parcel for SHAPESSOCIAL, the shared stimulus-locked signal available to any encoding model; it concentrates in superior temporal cortex.

3.2 A FEATURE BANK CHOSEN WITHOUT LOOKING AT EITHER TARGET

The bank must be selected before either target or the second description is examined. Within each sampled layer, features active in fewer than five or more than 95% of source-story TRs are excluded; the rest are ranked by binary temporal variance $p_k(1-p_k)$, where p_k is the fraction of source TRs in which the feature fires (Appendix E). Features are allocated approximately equally across the six layers, producing nested banks $K \in \{64, 128, 256, 512, 1024\}$ with $K = 256$ as the reference size. The identities are then frozen and carried to both targets and to the second description unchanged. For within-story analyses, selection is repeated inside each outer training fold. A correlation-pruned bank and correlated-feature clusters (complete linkage, $|\rho| \geq .80$) are built from the same source data for the redundancy analyses.

3.3 TWO MAPPINGS ON ONE BASIS

Both targets are predicted by multi-target ridge regression (Hoerl and Kennard, 1970) on the same coordinates but with different temporal assumptions. BOLD responds slowly, so its design stacks the current SAE vector and four preceding vectors (lags 0–4, spanning 0–6 s before the response). The model target is built from tokens in the same TR and uses the unlagged design. Both mappings use five contiguous outer temporal blocks with nested penalty selection, training-only scaling, and a four-TR purge around each BOLD test block so no lagged predictor of a test response appears in training (Varoquaux et al., 2017) (Appendices F and G).

We report Pearson r, which asks whether predicted and observed trajectories rise and fall together, and R^2_{oos}, which asks whether predictions reduce squared error relative to the training-target mean. The ridge penalty is selected separately for each score. An enhanced baseline with lexical, acoustic, positional, and surprisal covariates,

fit as a multiband ridge (Nunez-Elizalde et al., 2019; Dupré la Tour et al., 2022), tests whether the SAE block adds information beyond simpler stimulus features (Appendix H).

3.4 HELD-OUT RELIANCE

The central quantity is the change in test performance when a coordinate is removed while all fitted weights stay fixed. Feature k is set to its training mean in held-out data only; for BOLD, all five lagged columns are removed together. With $\widehat{Y}$ the ordinary prediction and $\widehat{Y}^{(-k)}$ the ablated prediction,

$$I_k \;=\; R^2_{\text{oos}}\big(Y, \widehat{Y}\big) \;-\; R^2_{\text{oos}}\big(Y, \widehat{Y}^{(-k)}\big). \tag{1}$$

Positive I_k means the mapping depends on the coordinate once the others are available; values near zero mean it does not. Losses are aggregated over target dimensions and, for BOLD, over participants. The primary correspondence statistic is the Spearman correlation between the BOLD and model reliance profiles. Grouped reliance repeats the procedure removing every member of a source-defined correlation cluster at once.

3.5 TRANSFER, NULLS, AND WHAT WOULD COUNT AS CORRESPONDENCE

Transfer. The decisive test transfers the source-story weights and scaling statistics to the second description without refitting, in both directions (SHAPESPHYSICAL→SHAPESSOCIAL and SHAPESSOCIAL→SHAPESPHYSICAL).

Temporal null. A positive profile correlation is not yet evidence, because both mappings inherit the story's temporal structure. The temporal null circularly shifts the model target by at least ten TRs, reruns nested penalty selection, refits the model mapping, recomputes reliance, and correlates the result with the fixed BOLD profile, for 199 shifts. The null is thus allowed to adapt fully to the wrong timing. Global null statistics synchronize the shift index across the twelve conditions before averaging, preserving dependence among them. With 199 shifts the smallest attainable one-sided p is $.005$.

Identity null. A second null permutes feature labels within each layer for 10,000 draws, preserving layer composition while breaking exact pairing. Passing it shows that exact identities carry information beyond layer membership; it is weaker than the temporal null because a statistic can depend on exact labels and still be explained by shared timing.

Interpretation criteria. We treat correspondence as established only if the global cross-story statistic beats both nulls, is positive in both directions, holds across three adjacent bank sizes including 256, appears in at least two model families after correction, and agrees between the variance-ranked bank and either the prevalence-ranked or pruned bank. Survival only after grouped ablation would indicate redundancy-dependent correspondence. Conversely, a null result is interpreted as *absence* only if the pipeline demonstrably recovers an injected correspondence of $\rho = .40$ with at least 80% power in at least two model families. The calibration (Appendix L) shows the design falls well short of that bar at $\rho = .40$, so the paper reports bounds rather than absence; Section 4.5 states exactly what the measured sensitivity licenses.

4 RESULTS

4.1 THE SOURCE-SELECTED BASIS PREDICTS BOTH TARGETS

Comparing feature losses is meaningful only if both targets are predictable from the transferred basis. They are, unequally. At $K = 256$ the BOLD mappings reached mean parcelwise r of .068–.092 and mean R^2_{oos} of .0034–.0084 across the twelve model–direction conditions; the paired model-state mappings reached r of .302–.475 and R^2_{oos} of .134–.271. The predictive signal is anatomically specific rather than diffuse: it concentrates in the same

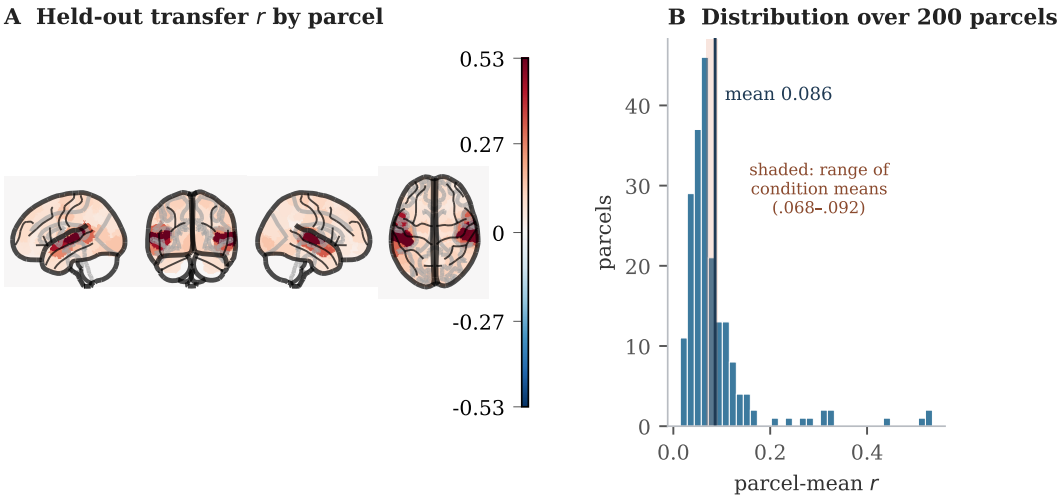


Figure 3: Where the transferred basis predicts BOLD. **A:** Parcelwise held-out transfer correlation, averaged over 59 participants, for Gemma 2 9B SHAPESPHYSICAL→SHAPESSOCIAL at $K = 256$; prediction concentrates in superior temporal and adjacent parcels, mirroring the ISC map in Fig. 2C. **B:** The distribution over parcels; the grand mean (.086) sits inside the range of the twelve condition means, so this condition is representative, and the long right tail shows that a few auditory parcels carry most of the signal.

superior temporal parcels that carry the shared stimulus-locked response (Fig. 3, Fig. 2C). The low BOLD values matter for everything that follows: explaining under one percent of held-out variance leaves little signal from which to estimate hundreds of small ablation losses.

The SAE block also adds only a little beyond simpler stimulus features. Within a description, adding it to the lexical, acoustic, positional, and surprisal baseline raised mean R^2_{oos} by .0010 for SHAPESPHYSICAL (95% bootstrap interval $[.0003, .0018]$, one-sided $p = .007$) and by .0034 for SHAPESSOCIAL ($[.0022, .0046]$, $p < .001$); the corresponding changes in r were $-.0001$ and .0047. In cross-description transfer, adding the block *reduced* mean r in both directions ($-.0118$ and $-.0108$; Appendix H). The basis is predictive, then, and carries a small amount of within-description variance beyond simple covariates, but that advantage does not transfer. This is enough signal to run the correspondence test; it is not itself a correspondence result.

4.2 BOLD AND THE MODEL DO NOT SHARE A RELIANCE ORDERING

At $K = 256$ the twelve directional Spearman correlations between conditional reliance profiles ranged from $-.626$ to .335 with mean $-.056$ (Table 1, Fig. 5B). Positive values in some conditions were offset by negative values in others, and neither sign nor rank order was stable across model family or transfer direction.

One cell looks favorable in isolation. Gemma 2 9B SHAPESPHYSICAL→SHAPESSOCIAL reached $\rho = .335$ with nominal temporal-null $p = .015$. Figure 4 shows this cell against its reverse direction: even the most favorable condition is a diffuse cloud in rank space. Its reverse direction was negative ($\rho = -.067$, $p = .805$), and its $p = .015$ became $q = .180$ after Benjamini–Hochberg correction across the twelve cells. Across all 120 direct cells in the bank-size sweep, seven had uncorrected $p < .05$, which is the number expected under the null; none survived correction. The pattern is heterogeneous, not reproducibly positive.

Redundancy does not explain the instability. Removing source-defined correlation clusters together gave twelve grouped correlations from $-.613$ to .303, mean $-.053$. The $K = 256$ banks retained 239–254 clusters (median 247), so grouping reduced effective dimension only modestly at the prespecified threshold; it changes the unit from a coordinate to a correlated family and reaches the same answer.

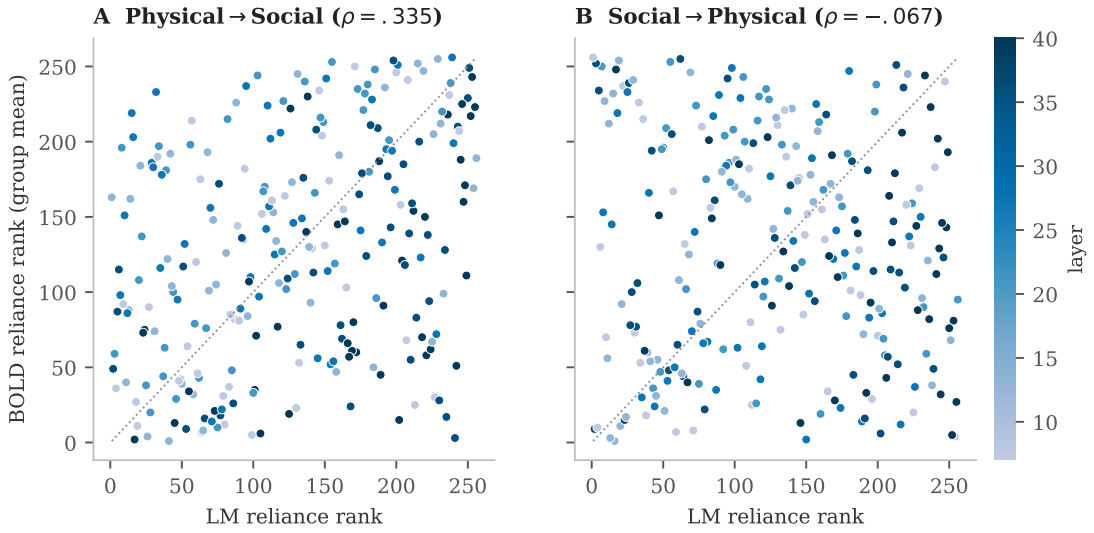


Figure 4: What the correspondence statistic actually compares: the same 256 coordinates ranked by group-mean BOLD reliance (vertical) and by LM reliance (horizontal), colored by source layer, for Gemma 2 9B at $K = 256$. **A:** the study's single most favorable condition ($\rho = .335$). **B:** the same model in the reverse direction ($\rho = -.067$). Even the best cell is a diffuse cloud rather than a ridge, and its structure does not survive reversing the transfer direction.

Table 1: Cross-story reliance correspondence across six models and both directions at $K = 256$. Ranges show every condition rather than a selected cell. p_{temporal} compares the observed mean Spearman correlation with 199 synchronized, fully retuned null means.

Analysis	Range of ρ	Mean ρ	p_{temporal}
Conditional, variance bank	$[-.626,\ .335]$	$-.056$	.890
Grouped, threshold .80	$[-.613,\ .303]$	$-.053$	.875
Conditional, pruned .80 bank	$[-.616,\ .286]$	$-.055$	.900

4.3 The ordering does not beat the temporal null

The observed mean is interpretable only against a null that keeps the story's structure. The synchronized conditional null means were centered near zero (mean .006, SD .050, 2.5th–97.5th percentiles $[-.101, .097]$), and 177 of the 199 equalled or exceeded the observed $-.056$, giving $p = (177 + 1)/(199 + 1) = .890$ (Fig. 5A). Grouped reliance gave $p = .875$. Each shift repeated penalty selection, fitting, and ablation, so the null was free to adapt to the wrong target timing; the observed ordering does no better than orderings fitted to misaligned targets.

The within-layer identity null gives only local evidence. Six of the twelve $K = 256$ cells had raw identity-null $p < .05$, so exact labels are not interchangeable in every condition. But only Gemma 2 9B ShapesPhysical→ShapesSocial cleared both nulls ($\rho = .335$, $p_{\text{identity}} < .001$, $p_{\text{temporal}} = .015$), its reverse direction did not ($\rho = -.067$, $p_{\text{identity}} = .836$, $p_{\text{temporal}} = .805$), and no other model family repeated it. Identity evidence in one cell does not rescue the global result.

4.4 No analysis choice we tested reverses the result

Bank size and pruning. Across five nested banks the all-condition mean stayed negative at every resolution (range $-.112$ to $-.056$; Fig. 5C, Table 8). A correlation-pruned bank averaged $-.055$. Increasing resolution reveals no trend toward positive correspondence, and selecting one representative per cluster does not recover one.

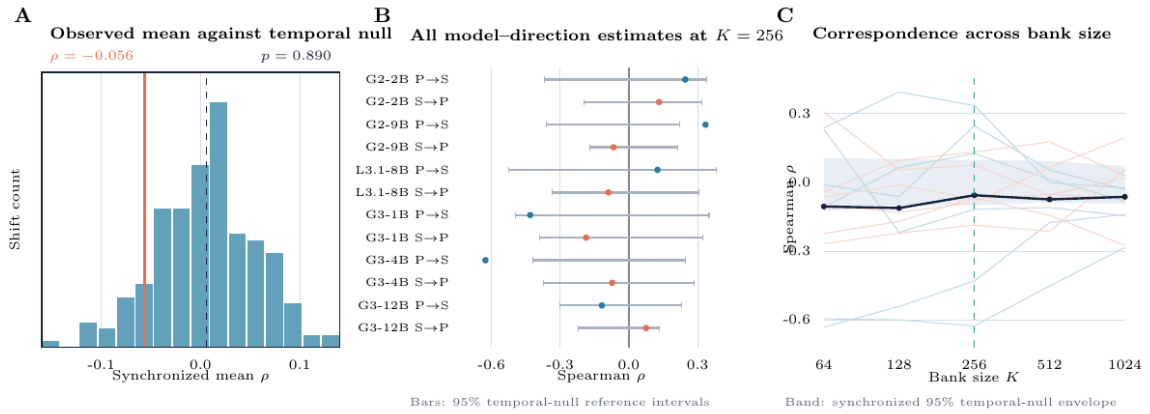


Figure 5: Observed correspondence against fully retuned temporal nulls. **A:** Distribution of 199 synchronized all-condition null means at $K = 256$; orange line, observed mean; dashed line, null mean. **B:** All twelve model–direction point estimates at $K = 256$; bars are condition-specific 2.5th–97.5th percentile temporal-null reference intervals, not confidence intervals. **C:** Observed all-condition mean across the nested bank-size sweep with the synchronized 95% null envelope and every fixed-order model–direction trajectory. Blue, SHAPESPHYSICAL→SHAPESSOCIAL; orange, SHAPESSOCIAL→SHAPESPHYSICAL; dashed vertical line, reference bank.

Token pooling. Mean, maximum, summed, and event-count pooling of token activations within a TR gave conditional means between $-.207$ and $-.056$, with grouped counterparts from $-.206$ to $-.053$. No pooling rule made the mean positive, so dilution of brief activations by mean pooling does not explain the result.

The negative sign is not a finding. The mean is negative under every setting, which invites an anti-correspondence reading. It does not support one. Removing Gemma 3 4B alone moves the primary mean to $+.002$; removing Gemma 3 1B leaves it at $-.006$. Sum and event-count pooling are more negative, but they weight feature frequency differently and were not designed as anti-correspondence tests. The pattern motivates a targeted diagnostic of prevalence, collinearity, and ridge shrinkage; it does not establish an inverse organization.

Noise and averaging. Median parcelwise leave-one-participant-out inter-subject correlation was .162 for SHAPESPHYSICAL and .197 for SHAPESSOCIAL; split-half group correlations were .542 and .612 (Table 7). Within-story reliance reliability was only .074, whereas cross-story participant reliability was .289 and split-half group reliability .745. These come from different estimators: within-story profiles are rebuilt from five shorter folds with fold-specific banks, whereas transfer uses one source bank and all 303–305 target TRs, so the higher transfer value does not show that transfer is intrinsically more reliable. Averaging uncensored responses across participants raised mean-fold BOLD r from .022–.064 to .166–.303, yet the resulting reliance correlations still ranged from $-.164$ to .535. Averaging improves prediction; it does not produce a consistently positive correspondence.

Together these checks rule out the immediate alternative explanations. The conclusion is not created by one bank size, one pooling rule, ungrouped redundancy, or participant noise alone. They also show why favorable individual cells are fragile: the effective feature dimension is smaller than the nominal bank, and participant-level reliance is noisy.

4.5 MEASURED DETECTABILITY: WHAT THE NULL CAN AND CANNOT MEAN

The completed analyses do not support a robust positive interpretation. Whether they support an absence interpretation depends on what the pipeline could have detected, and we measured that directly for one family. A semi-synthetic calibration injects a known latent reliance correspondence at the 256-feature group level, simulates each target from its own observed predictor geometry and empirical residual pool, and runs the complete fitting,

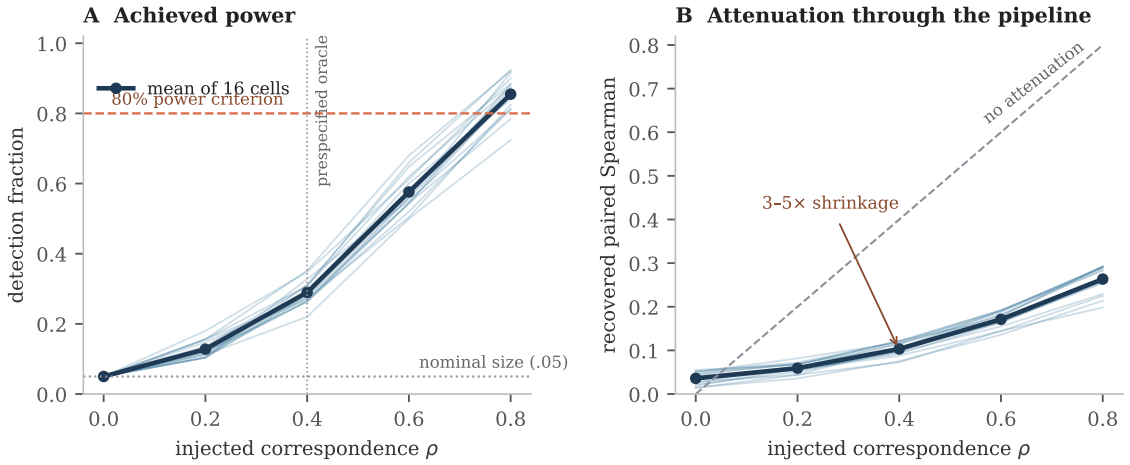


Figure 6: What the pipeline can see. **A:** Detection fraction versus injected correspondence for all sixteen Gemma 2 2B calibration cells (light lines) and their mean (dark); the test holds its nominal .05 size at $\rho = 0$ but crosses 80% power only near $\rho = .8$, and at the prespecified oracle $\rho = .4$ detects .22–.35 of injections. **B:** Recovered paired Spearman versus injected ρ; the estimation chain shrinks the injected effect three- to five-fold, which is why moderate correspondence falls below the noise threshold.

Table 2: Semi-synthetic detectability for Gemma 2 2B. Each oracle row summarizes sixteen cells: four designs (variance and pruned .80 banks $\times$ both transfer directions) crossed with residual-scale multipliers $\{0.25, 0.5, 1, 2\}$; 250 positive replicates per cell, 500 $\rho = 0$ replicates per design. Detection means the paired held-out reliance Spearman exceeds the 95th percentile of the same-design $\rho = 0$ distribution.

Oracle ρ	Detection range	Cells $\geq$ 80% power	Mean recovered ρ
.0	.050 (all cells)	0/16	.014–.055
.2	.104–.180	0/16	.036–.082
.4	.220–.352	0/16	.073–.122
.6	.500–.680	0/16	.136–.192
.8	.724–.924	14/16	.198–.293

transfer, ablation, and null machinery on the result (Appendix L). Four Gemma 2 2B designs were computed (variance and pruned banks, both directions), each over oracle correlations $\{0, .2, .4, .6, .8\}$ and four residual scales, with 500 null and 250 positive replicates per cell.

The calibration is well behaved and sobering (Fig. 6, Table 2). At $\rho = 0$ the detection rate is exactly the nominal .050 in all sixteen cells, so the test is not anticonservative. At the prespecified oracle $\rho = .40$, detection is .22–.35 — far below the 80% criterion in every cell. Detection reaches .50–.68 at $\rho = .60$ and exceeds 80% only near $\rho = .80$ (14 of 16 cells). The attenuation column explains why: an injected $\rho = .40$ emerges from fitting, ablation, and aggregation as a measured paired correlation of only .07–.12, a three- to five-fold shrinkage produced by the estimation chain itself, against single-design 95% noise thresholds of .13–.17.

The negative result is therefore bounded in a specific, quantified way. For Gemma 2 2B, a strong correspondence ($\rho \approx .80$) would likely have been detected and was not observed, so the data weigh against it; a moderate correspondence ($\rho \leq .60$) would more often than not have been missed, so the data cannot distinguish it from absence. These bounds are measured directly for Gemma 2 2B and untested for the other five systems, which share the design but not necessarily its noise geometry. The conclusion is unchanged in kind and sharper in degree: the study does not establish sparse feature correspondence, and it bounds, rather than excludes, its presence.

5 DISCUSSION

The experiment separates predictive overlap from shared organization. A source-selected SAE basis predicts both targets on a second description, so the comparison is not run on two arbitrary noise fields. Yet the BOLD and model mappings do not agree on which coordinates matter: their mean conditional and grouped correlations are slightly negative, neither produces bidirectionally replicated temporal-null evidence, and neither becomes positive under the redundancy, resolution, pooling, or averaging checks. Sparse feature correspondence is not established by this experiment.

This is why encoding accuracy and correspondence must be reported separately. Prediction asks whether a representation contains information that supports a target. Reliance correspondence asks whether two fitted mappings depend on the same coordinates in the same order. A shared basis can succeed at the first because both targets follow the stimulus, and fail at the second because the mappings use different combinations of correlated features. Sparse coordinates make that distinction testable; they do not erase it.

The two nulls sharpen the interpretation at different scopes. Several cells survive within-layer relabeling, so exact labels can matter in an individual condition. Only one also clears its temporal null, and it does not replicate in the reverse direction or across model families. At the global level, the observed ordering does not outperform models fit after the model target is moved to the wrong narrative time. Local identity evidence is therefore not evidence for shared feature organization.

The model target sets the scope of the conclusion. Predicting the final hidden state tests whether earlier-layer SAE activations are linearly recoverable in that surrogate; it is not the same as asking which coordinates change next-token loss or behavior. The present result does not establish robust correspondence for this final-state surrogate and cannot rule out a different ordering for a behavior-facing target. A direct follow-up would keep the same source-only selection, transfer, and null logic while measuring held-out change in next-token loss or distributional KL divergence.

The calibration converts the decisive uncertainty into a measured limit. The pipeline's false-positive rate is nominal, but its sensitivity is not adequate for the effects most plausibly at stake: a moderate injected correspondence is shrunk three- to five-fold by the estimation chain and detected less than a third of the time. Two consequences follow. First, the observed null is informative about strong correspondence only — for the calibrated family, $\rho \approx .80$ would likely have been seen — and says little about moderate correspondence. Second, the path to a decisive experiment is now concrete: the attenuation, not the null thresholds, is the binding constraint, so progress requires more independent stories, denser sampling of participants per story, or a reliance estimator with less shrinkage, before more model families are added. Only after a feature replicates across stories and dictionaries, survives matched negative and counterfactual texts, and defines an outcome-blind BOLD contrast should it be interpreted at the feature level.

6 LIMITATIONS

The two descriptions change words and timing but describe one animation, so the transfer test is across descriptions of shared events, not across an independent stimulus population. BOLD is an indirect vascular signal sampled every 1.5 s and summarized in 200 parcels; the lagged ridge model approximates delayed responses and is not a model of neuronal computation, and group reliability does not provide an acquisition-level noise ceiling. The model target is a linear surrogate for the final hidden state and may emphasize architectural propagation over features that matter to behavior. An SAE basis is not canonical: feature identities depend on model, layer, dictionary training, width, and seed (Leask et al., 2025; Paulo and Belrose, 2026), and grouping and pruning test specific forms of redundancy without creating biological units. The detectability calibration covers one model family; its attenuation and power estimates should generalize only insofar as the other systems share Gemma 2 2B's predictor geometry and noise structure, which is untested. For these reasons the paper makes no claim about neurons, cortical mechanisms, linguistic causation, subcortical systems, or auditory regions.

7 CONCLUSION

Across six model–SAE systems, a sparse basis selected from one description of a story predicts both BOLD and the model's own state on a second description. It does not produce a reliable shared ordering of feature importance: conditional and grouped reliance provide no bidirectionally replicated temporal-null evidence, and the result survives every sensitivity check we ran. A semi-synthetic calibration on one family shows why the negative claim must stay bounded: the pipeline's size is exact, but an injected moderate correspondence is attenuated below its own noise threshold and detected a minority of the time, with 80% power only near $\rho = .80$. The experiment therefore weighs against strong correspondence in the calibrated family and cannot decide between moderate correspondence and absence. Prediction starts the comparison; transfer, structured nulls, and calibrated recovery decide what it means.

REPRODUCIBILITY STATEMENT

A code release accompanies the submission: the complete scientific workflow (feature-bank construction, both fitting regimes, transfer, conditional and grouped ablation, both null generators, pooling and reliability sensitivities, the semi-synthetic power stage, figure builders, and the artifact verifier), a SHA-256 checksum catalog covering every source file, a pinned dependency list, and a data-availability statement naming the external inputs reproduction requires (BOLD target bundles, predictor and target bundles, transcript timing assets, model and SAE checkpoints). Raw fMRI data, participant identifiers, stimuli, and model weights are not redistributed; paths bind to an authorized local mirror through documented environment variables, and inputs are accepted by content hash, not filename. Each result artifact logs its code hash, configuration fingerprint, model, story, participant, fold, feature bank, selected penalty, reliance profile, and null values; the cache rejects any artifact whose requested fingerprint differs. All reported values were recomputed from these condition-level artifacts, and the detectability stage ships 128 hash-verified shards with four single-design aggregates for the calibrated family.

USE OF LARGE LANGUAGE MODELS

AI assistants were used during this work as tools: for implementation support in the analysis and infrastructure code, and for rephrasing and editing prose in the manuscript. All scientific decisions, analyses, verification of results, and the final text are the authors' own, and the authors take full responsibility for the correctness of the code, the reported numbers, and every claim in the paper.

ETHICS STATEMENT

This is a secondary analysis of the publicly released Narratives dataset; no new human-subject data were collected. Consent and ethics procedures for the source dataset are described by Nastase et al. (2021). The analysis uses deidentified parcel-level derivatives and reports aggregate results only. BOLD is treated as a parcellated measurement and SAE coordinates as model-analysis variables, not as neurons, cortical mechanisms, clinical biomarkers, diagnoses, or individual assessments.

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A NARRATIVE DATA AND COHORT CONSTRUCTION

The analysis uses the Narratives naturalistic-language fMRI dataset (OpenNeuro ds002345; Nastase et al., 2021). Participants listened to continuous spoken narratives during fMRI acquisition. The evaluation stories are SHAPESPHYSICAL and SHAPESSOCIAL, which share TR, scanner family, and closely matched acquisition settings and are alternative verbal descriptions of the same animation. Bidirectional transfer therefore probes a specific description shift under a shared visual event sequence, not a broad sample of narrative domains.

Table 3: Evaluation cohorts. Full rows equal participant-runs times story TRs. Flagged rows remain on the saved timing grid but are excluded as response rows during fitting.

Story	TR (s)	TRs	Runs	Full rows	Flagged	Retained
SHAPESPHYSICAL	1.5	303	59	17,877	487	17,390
SHAPESSOCIAL	1.5	305	59	17,995	488	17,507

Every primary target has 200 cortical parcels. The full-59 SHAPESPHYSICAL target was assembled from 47 previously completed derivatives and 12 independently recovered derivatives; assembly required a unique participant/run key, the expected 303-row grid for every run, identical parcel labels, finite arrays, and matching per-file hashes, and the final target was independently checked before primary fitting. Earlier 48-run SHAPESPHYSICAL analyses and the matched 48-run SHAPESSOCIAL subset appear only as cohort sensitivities; every primary number uses all 59 eligible participant-runs. Participant-level arrays and the full 59-run provenance table ship with the release.

The target is a parcellated BOLD response, not neuronal firing and not single-neuron activity. BOLD reflects vascular changes coupled to aggregate neural activity and is temporally blurred relative to the underlying events (Chen and Glover, 2015). All claims are accordingly phrased as prediction of BOLD responses.

B fMRI PREPROCESSING AND TARGET CONSTRUCTION

Extraction. The pipeline starts from fMRIPrep derivatives in `MNI152NLin6Asym` space (Esteban et al., 2019). For each participant-run, `NiftiLabelsMasker` samples the Schaefer-2018 200-parcel, seven-network cortical atlas (Abraham et al., 2014; Schaefer et al., 2018) with `standardize=False`, `detrend=False`, no smoothing, and no temporal filters, so that extraction and cleaning are separate, auditable operations. Parcel labels are intersected across retained runs; all 200 survive in both bundles.

Nuisance regression and filtering. The nuisance design contains 24 motion regressors (six rigid-body estimates, their temporal derivatives, and the squares of both) and the first six anatomical CompCor components (Behzadi et al., 2007). Time series are cleaned with detrending, standardized confounds, and a 0.008-Hz high-pass filter; no low-pass filter and no global-signal regressor are used. Runs are then parcel-standardized over the uncensored TR set U_s:

$$z_{s,t,p} = \frac{y_{s,t,p} - \mu^{(U)}_{s,p}}{\sigma^{(U)}_{s,p}}, \qquad \mu^{(U)}_{s,p} = \frac{1}{|U_s|} \sum_{t \in U_s} y_{s,t,p}, \tag{2}$$

with $\sigma^{(U)}$ likewise computed over U_s. This run-wise target construction is distinct from the training-fold scaling used by ridge regression (Appendix F).

Censoring. A response TR is flagged when framewise displacement exceeds 0.5 mm, standardized DVARS exceeds 1.5, or an fMRIPrep non-steady-state indicator is active. The displacement and DVARS diagnostics follow Power et al. (2012) and Afyouni and Nichols (2018); the 0.5 mm and 1.5 cutoffs are prespecified analysis thresholds of this study, not values those papers prescribe. The full TR grid and flag vector are preserved; flagged rows are omitted only when a design matrix is built, so word/token/TR alignment is preserved and earlier uncensored lags

remain available as predictors for a later valid response. No censored target contributes to loss, validation, or test metrics.

C TRANSCRIPT, TOKEN, AND TR ALIGNMENT

Timing. Word timings were generated in stored 60-s audio chunks and stitched to the story clock; the saved stimulus-onset correction is applied before TR assignment. A word with onset o_i and offset d_i is assigned by midpoint,

$$t_i = \left\lfloor \frac{(o_i + d_i)/2 - \delta_{\text{onset}}}{1.5} \right\rfloor. \tag{3}$$

Out-of-range indices trigger a validation failure rather than clipping into an endpoint bin.

Tokenization. The transcript is tokenized separately for every base model without added special tokens. Extraction-window records store global token indices; the pipeline requires these to be contiguous, reconstructs a single story token stream, groups subword tokens back to words, and checks normalized text and word count against the timing assets. The reconstructed words in every TR are compared with the saved TR-aligned transcript; any mismatch stops the analysis.

TR aggregation. For selected feature k, the primary TR predictor is the mean of its stored positive token activations in that bin,

$$x_{t,k} = \begin{cases} |A_{t,k}|^{-1} \sum_{i \in A_{t,k}} a_{i,k}, & |A_{t,k}| > 0, \\ 0, & |A_{t,k}| = 0, \end{cases} \tag{4}$$

where zero denotes no recorded activation in the bin. Presence, summed mass, peak, and count views are retained as pooling sensitivities. Missing pre-story lag vectors are zero-filled; observed bins are never copied forward or interpolated.

D MODELS, SAE RELEASES, AND EXTRACTED TARGETS

Gemma Scope and Llama Scope provide the model-matched residual-stream SAE dictionaries (Gemma Team, 2024; Grattafiori et al., 2024; Lieberum et al., 2024; He et al., 2024); the Gemma 3 systems use the listed checkpoints (Google DeepMind, 2025a; Gemma Team, 2025) and the Gemma Scope 2 dictionaries (Google DeepMind, 2025b). SAE identities are local to a model, layer, release, and dictionary; a feature index is never treated as shared across base models.

Table 4: Model and SAE configurations. Layer indices are the six depth-spaced residual-stream sites used for each model-specific feature bank.

Model	Model ID	SAE release	Dict. size/layer	Layers
Gemma 2 2B	`google/gemma-2-2b`	Gemma Scope PT residual canonical	16,384	4, 8, 13, 17, 22, 25
Gemma 2 9B	`google/gemma-2-9b`	Gemma Scope PT residual canonical	16,384	7, 13, 21, 27, 36, 40
Llama 3.1 8B	`meta-llama/Llama-3.1-8B`	Llama Scope LXR	32,768	5, 10, 16, 21, 27, 31
Gemma 3 1B	`google/gemma-3-1b-pt`	Gemma Scope 2 residual, big-L_0	16,384	4, 8, 13, 17, 22, 25
Gemma 3 4B	`google/gemma-3-4b-pt`	Gemma Scope 2 residual, big-L_0	16,384	5, 10, 17, 22, 29, 33
Gemma 3 12B	`google/gemma-3-12b-pt`	Gemma Scope 2 residual, big-L_0	16,384	7, 15, 24, 31, 41, 47

The canonical model target at TR t is the mean final hidden state before the LM head over the tokens assigned to that TR,

$$\boldsymbol{h}_t = |B_t|^{-1} \sum_{i \in B_t} \boldsymbol{h}_i^{\text{final}}. \tag{5}$$

Extraction uses consecutive causal windows of at most 1,024 tokens; context is available only within the current window, and no result is described as full-transcript KV-cache extraction. Empty target bins use the stored zero-vector convention and are recorded in the artifact metadata.

E DIRECTION-SPECIFIC SOURCE-STORY FEATURE BANKS

Exclusion firewall. The primary bank is selected from the training-side stimulus only. For SHAPESPHYSICAL→SHAPESSOCIAL, selection uses SHAPESPHYSICAL predictors and is completed before SHAPESSOCIAL is examined; the reverse direction uses a separately selected SHAPESSOCIAL bank. BOLD values, model targets, fitted weights, fold scores, null statistics, and all target-story predictors are excluded from selection. Within-story temporal evaluation repeats selection inside each outer training fold. The same selected feature–layer identities are then used for all participants and for the model within that fold or direction.

Ranking. For feature k, let p_k be the fraction of eligible source TRs in which its positive activation is nonzero. Features active in fewer than five or more than 95% of source TRs are excluded. Within each sampled layer, eligible features are ranked by binary temporal variance $V_k = p_k(1 - p_k)$, with ties broken by positive-activation RMS and then feature ID. Features are allocated approximately equally across the six layers, producing nested prefixes $K \in \{64, 128, 256, 512, 1024\}$ with 256 as the primary size. A prevalence-only sensitivity ranks by p_k under the same filter. Correlation-cluster sensitivities use deterministic complete linkage within layer at absolute Spearman thresholds .70, .80, and .90, with the highest-ranked variance feature as representative and full membership preserved for grouped ablation.

Table 5: Realized source-bank coverage. Eligible-feature counts are listed in the layer order of Table 4. The last column gives the number of complete-linkage clusters in the $K = 256$, $|\rho| \geq .80$ bank.

Model	Source	Eligible features by layer	Clusters
Gemma 2 2B	Physical	3488/1874/2053/1624/1612/119	246
	Social	3981/2120/1969/1762/1959/115	254
Gemma 2 9B	Physical	2178/2643/1928/1811/938/1253	250
	Social	2567/3249/2221/1995/1024/1368	254
Gemma 3 1B	Physical	3840/4300/4315/3704/2365/45	239
	Social	4395/5088/5179/4787/3275/46	241
Gemma 3 4B	Physical	3248/3005/2507/3029/2847/34	248
	Social	3626/3386/2836/3814/3817/34	249
Gemma 3 12B	Physical	2865/2618/1832/1898/1939/32	244
	Social	3356/3009/1858/2127/2488/32	244
Llama 3.1 8B	Physical	1479/1373/1398/1250/1216/11805	240
	Social	1701/1507/1515/1627/1693/18553	249

The deepest Gemma 3 site contributes few eligible features (32–46), so at $K = 256$ that layer is represented by essentially all of its eligible coordinates and at larger K cannot fill an equal share; the remaining slots are filled from the other layers in rank order. A reciprocal-rank bank built from five other Narratives stories provides a sensitivity whose selection corpus excludes both Shapes stories. The selection procedure is outcome-blind but does not certify semantic quality: features can be correlated, polysemantic, dictionary-specific, or sensitive to activation prevalence (Cunningham et al., 2023; Leask et al., 2025).

F PREDICTION MODELS AND SCALING

Designs. For a bank of size K, the BOLD design at response TR t is

$$\widetilde{\boldsymbol{x}}_t = [\boldsymbol{x}_t, \boldsymbol{x}_{t-1}, \ldots, \boldsymbol{x}_{t-4}] \in \mathbb{R}^{5K}, \tag{6}$$

and all five columns of one SAE feature are treated as a single group in reliance analysis. The model design is the unlagged $\boldsymbol{x}_t \in \mathbb{R}^K$ because its target is aggregated from tokens in the same TR. For predictor matrix X, target matrix Y, and coefficient matrix B, ordinary multi-target ridge solves

$$\widehat{B} = \arg\min_B \|Y - XB\|_F^2 + \lambda \|B\|_F^2, \tag{7}$$

with candidate grid $\{10^{-1}, 10^0, \ldots, 10^{10}\}$. Pearson r and R^2_{oos} claims use separately selected penalties.

Training-only scaling. Every predictor and target dimension is standardized with the current training split's statistics, $X^* = (X - \mu_X^{\text{tr}})/\sigma_X^{\text{tr}}$, applied unchanged to validation and test inputs; any zero training scale is replaced by one. Targets are scaled analogously during fitting and predictions are returned to target units before scoring. Test rows never contribute to scaling or penalty selection.

Metrics. Pearson correlation is the primary predictive-shape score because its penalty is selected by validation correlation. The variance metric uses the training target mean as its out-of-sample baseline,

$$R^2_{\text{oos}} = 1 - \frac{\sum_{i \in \text{te}} (y_i - \widehat{y}_i)^2}{\sum_{i \in \text{te}} (y_i - \bar{y}_{\text{tr}})^2}. \tag{8}$$

Using the test mean instead would change the estimand.

G EVALUATION REGIMES

Participant-specific purged temporal evaluation. Each participant's BOLD mapping is evaluated in five contiguous outer temporal folds. Four adjacent response TRs are purged from training on each side of the test block; because the brain model uses lags 0–4, the nearest retained training response is five TRs from the held-out block and cannot contain one of its lagged predictor vectors. Nested inner folds repeat the purge and select the penalty, following blocked-validation practice for temporally dependent neuroimaging data (Varoquaux et al., 2017). This regime tests new time blocks within the same story for the same participant.

Cross-participant shared-stimulus bridge. Participants are divided into five held-out groups; four nested participant folds select the penalty. Every participant hears the same story at the same TR indices, so the predictor row repeats across participants. With n_t training responses at time t and mean $\bar{y}_t$, the repeated-row loss decomposes exactly as

$$\sum_{s=1}^{n_t} \|y_{s,t} - x_t B\|_2^2 = n_t \|\bar{y}_t - x_t B\|_2^2 + C_t, \tag{9}$$

with C_t independent of B, so the weighted reduction gives the same ridge solution as explicit row repetition (verified by a synthetic unit test). This regime tests new listeners at known stimulus times, not unseen text or time points.

Bidirectional cross-story transfer. Penalties and scaling statistics are learned on one story, the final mapping is fit on that story, and weights are applied unchanged to the other; the procedure is repeated in both directions. All 59 participants are available for both stories. This is the strictest current test of stimulus generalization, although it still contains only two stories.

H ENHANCED BASELINES AND MULTIBAND CONTROLS

The enhanced baseline separates three predictor blocks: (1) low-level and positional variables (word count, phoneme count, audio envelope level and change, story-position ramp/plateau bases); (2) model-specific mean and summed

token surprisal per TR; (3) the fixed SAE feature bank. The multiband model assigns an independent ridge penalty to every block (Nunez-Elizalde et al., 2019; Dupré la Tour et al., 2022),

$$\min_{B_1,B_2,B_3} \left\| Y - \sum_{b=1}^{3} X_b B_b \right\|_F^2 + \sum_{b=1}^{3} \lambda_b \|B_b\|_F^2, \tag{10}$$

with ratios spanning 10^{-4} to 10^4 plus a prespecified random-search candidate set. Dense residual-stream encodings at the corresponding layers and 99 matched-random SAE banks are representation controls. The participant-clustered comparison reuses the same participant folds throughout.

Table 6: Global participant-paired SAE-block increments over the enhanced baseline at $K = 256$. Intervals are 95% participant-bootstrap intervals.

Regime	Direction/story	Mean Δr	Mean ΔR^2_{oos}
Within	Physical	$-.0001$	.0010
Within	Social	.0047	.0034
Transfer	Physical→Social	$-.0118$	$-.0019$
Transfer	Social→Physical	$-.0108$	$-.0006$

The within-story ΔR^2_{oos} intervals exclude zero for both descriptions: $[.0003, .0018]$ for SHAPESPHYSICAL and $[.0022, .0046]$ for SHAPESSOCIAL. Both transfer Δr intervals are negative: $[-.0170, -.0069]$ and $[-.0163, -.0055]$. Dense-state and matched-random banks serve as representation controls for these increments; the source-story comparison above is the primary one.

I CONDITIONAL SINGLE-FEATURE AND GROUPED RELIANCE

Coefficient magnitude is not the feature-importance measure. For SAE feature k, all of its standardized BOLD test columns (lags 0–4) are set to zero, which equals the training mean after scaling; for the model mapping its one test column is set to zero. The fitted coefficients are not retrained. Reliance is the reduction in held-out R^2_{oos} (Eq. (1)). For transfer, scores are aggregated over target dimensions and then participants; within-story correlations are calculated per fold because fold-specific banks may contain different identities. The primary cross-target statistic is the Spearman correlation between paired BOLD and model reliance scores; Pearson correlation and matched top-12.5% overlap are sensitivities. For every source-defined correlation cluster, grouped reliance jointly zeros all member coordinates and, for BOLD, every lag of every member. If too few independent clusters remain, the grouped comparison is reported as insufficiently resolved rather than negative.

This measure asks whether removing the same standardized predictor group harms held-out prediction in both fitted mappings. It does not imply that the feature is causal, uniquely identifiable among correlated predictors, or a biological unit.

J NULLS, UNCERTAINTY, AND INTERPRETATION CRITERIA

Temporal-target null. The model target is circularly shifted in story time by at least ten TRs. For every shift, nested penalty selection is rerun, the model mapping is refit, held-out reliance is recomputed, and correspondence with the fixed BOLD reliance profile is evaluated. There are 199 deterministic shifts. Global null statistics synchronize the shift index across conditions before averaging, preserving dependence among model/story results. The temporal test is substantially more expensive than the identity test: each shift repeats selection, fitting, and ablation, whereas an identity draw permutes an already estimated profile. With 199 shifts the minimum attainable one-sided Monte Carlo p is .005.

Within-layer identity null. Model feature identities are permuted within sampled layer, not globally, for 10,000 draws. This preserves layer allocation and layer-level preference while testing exact feature identity. Passing it is weaker than passing the temporal null: a statistic can depend on exact labels and still be explained by shared temporal structure.

Uncertainty and multiplicity. BOLD prediction uncertainty uses fixed-seed participant bootstrap draws. Reliance stability is reported across held-out folds and participants. Participant-level model increments use paired sign-flip tests and bootstrap intervals. Benjamini–Hochberg correction is applied across the twelve model–story conditions, and a separate global mean-effect test prevents a count of positive point estimates from being interpreted as evidence.

Criteria. Robust correspondence requires the global cross-story statistic to beat both nulls, be positive in both directions, repeat across three adjacent K values including 256, appear in at least two model families after correction, and agree under the variance bank and either the prevalence or pruned bank. Survival only after grouped ablation indicates redundancy-dependent correspondence. A non-significant result is interpreted as an informative null only if conditional and grouped results remain null across bank size, selection, and pooling, with at least 80% achieved power for an oracle $\rho = .40$. Below that resolution the data do not distinguish a small effect from insufficient measurement sensitivity.

K SENSITIVITY ANALYSES AND MEASUREMENT DIAGNOSIS

The main result could fail because BOLD is noisy, the sparse bank is poorly conditioned, or an analysis choice suppresses brief activations. Each explanation was evaluated without changing the primary estimand.

Response and reliance reliability. Response consistency and feature-reliance stability are different quantities; Table 7 reports them separately. These are shared-response and derived-profile diagnostics, not an acquisition-level BOLD noise ceiling. The group-average sensitivity changes the target, not the predictors: averaging uncensored responses raised mean-fold BOLD r from .022–.064 to .166–.303, pooled out-of-fold r ranged from .054 to .210, and R^2_{oos} was positive in 11 of 12 conditions. Reliance correspondence nevertheless remained heterogeneous ($-.164$ to .535).

Table 7: Reliability diagnostics. LOO-ISC is median parcelwise leave-one-participant-out response correlation. Split-half values are median parcelwise correlations across 10,000 participant splits, followed by their Spearman–Brown correction. Reliance values are median exact-coordinate Spearman correlations for R^2_{oos} profiles at $K = 256$. None is an acquisition-level noise ceiling.

Quantity	Evaluation	Individual/LOO	Split-half group	Corrected
BOLD response	Physical	.162	.542	.703
BOLD response	Social	.197	.612	.759
BOLD reliance	Within story	.074	.369	–
BOLD reliance	Cross-story transfer	.289	.745	–

Coverage, redundancy, and pooling. The conditioning audit records prevalence, zero fraction, temporal variance, positive RMS, rare-feature counts, pairwise absolute correlations, rank, condition number, cluster structure, source-to-target shift, and directional bank overlap; fold-specific quantities use training data only. The primary bank, grouped clusters, and pruned representatives answer different questions but all fail the temporal criterion, and mean, maximum, sum, and event-count pooling leave the global conclusion unchanged (Table 8); the occasional favorable cell does not replicate bidirectionally.

Table 8: All-condition mean cross-story conditional reliance correspondence under bank-size, pruning, and token-pooling sensitivities. Every row averages the same six models and two directions; none selects a favorable cell. Temporal values use synchronized Spearman-null draws with denominator 200.

Sensitivity	Setting	Mean ρ	p_{temporal}
Bank size	$K = 64$	$-.106$	.950
	$K = 128$	$-.112$	.990
	$K = 256$	$-.056$	.890
	$K = 512$	$-.075$	.910
	$K = 1024$	$-.063$	.890
Pruning	threshold .80, $K = 256$	$-.055$	.900
Pooling	mean positive activation	$-.056$	.720
	maximum positive activation	$-.102$	.910
	summed positive activation	$-.189$	.970
	event count	$-.207$	.965

L SEMI-SYNTHETIC DETECTABILITY CALIBRATION

Paired-design construction. One paired design is materialized per model, bank family, and direction. Each design records the target-story SAE predictor values under the source-story bank; the five lag-expanded BOLD columns per feature (1,280 columns at $K = 256$) and the unlagged model columns (256) as distinct matrices; participant-major BOLD rows and TR-level model rows with their own fold structures (participant-by-time blocks for BOLD, contiguous TR blocks for the model target); empirical residual pools drawn from the corrected R^2_{oos}-selected transfer predictions with censored BOLD rows excluded; and censor masks with full metadata and code/input hashes. The lagged BOLD matrix and the model matrix are never treated as the same predictor space: the same latent reliance profile is injected at the 256 SAE-feature *group* level, while each target is simulated from its own observed geometry and empirical residual covariance, at residual-scale multipliers $\{0.25, 0.5, 1, 2\}$.

Decision rule and replication. For each cell, oracle correlations $\{0, .2, .4, .6, .8\}$ are injected with 250 positive replicates, and 500 replicates generate the same-design $\rho = 0$ distribution. A replicate counts as detected when its paired held-out reliance Spearman exceeds that distribution's 95th percentile; a design is identifiable at an oracle when the detection fraction reaches .80. Replicate seeds derive from the base seed, design SHA-256, oracle correlation, residual scale, and replicate ID, so results are independent of execution order. Every shard is hash-verified against a sidecar before aggregation, and aggregation refuses partial designs.

Coverage and outcome. The four Gemma 2 2B designs (variance and pruned .80 banks, both directions; 128 verified shards) are complete; Table 2 summarizes them. Detection at $\rho = 0$ is .050 in every cell; achieved power at the prespecified oracle $\rho = .40$ is .22–.35; the .80 criterion is reached only near $\rho = .80$ (14/16 cells, the exceptions at the smallest residual scale). Recovered mean correlations show three- to five-fold attenuation of the injected effect. The calibration covers Gemma 2 2B; the corresponding Llama 3.1 8B designs were not computed. Its sensitivity estimates are therefore direct for one model family and untested for the rest, which is why the paper reports bounds rather than absence.

M COEFFICIENT DIAGNOSTICS

For standardized coefficient energy, BOLD coefficients are indexed by feature k, lag ℓ, and parcel p. Two aggregations are retained:

$$E_k^{\text{signed}} = \Big\| \sum_{\ell=0}^{4} \beta_{k,\ell,:} \Big\|_2, \qquad E_k^{\text{lag}} = \Big(\sum_{\ell=0}^{4} \|\beta_{k,\ell,:}\|_2^2 \Big)^{1/2}. \tag{11}$$

The first permits opposing lag weights to cancel; the second does not. Model-side energy is $\|\gamma_{k,:}\|_2$. These summaries are called "coefficient-energy correspondence," not feature importance. Raw-unit and standardized-unit results are reported together to show how activation scale changes apparent agreement.

N EXPLORATORY RSA AND ANATOMICAL SUMMARIES

Representational similarity analysis compares relational geometry through representational dissimilarity matrices (Kriegeskorte et al., 2008). With $S^{\mathrm{B}}_{\mathrm{OOF}}$ and $S^{\mathrm{LM}}_{\mathrm{OOF}}$ constructed only from out-of-fold predictions, their upper triangles are compared as $\mathrm{RSA} = \mathrm{corr}(\mathrm{vec}_u(S^{\mathrm{B}}_{\mathrm{OOF}}), \mathrm{vec}_u(S^{\mathrm{LM}}_{\mathrm{OOF}}))$. Every null must repeat target shifting, penalty selection, fitting, and matrix construction. RSA remains exploratory; a non-significant result is a failure to establish geometry correspondence, not evidence that the geometries differ. Parcelwise maps and network summaries are likewise exploratory: they may show where a stimulus model predicts BOLD most strongly, but anatomical labels are not used to infer that an SAE feature is located in a parcel.

O FEATURE INTERPRETATION AND PROSPECTIVE PROBING

Dolma contexts, automated conceptual labels, correlated-feature comparisons, and localized deletion tests were assembled to audit candidate SAE features. These materials are kept separate from feature selection and primary inference. A complete empirical probe would require multiple natural positive examples; matched natural negative examples; a minimal counterfactual that changes the proposed property while holding lexical, length, and formatting cues fixed; a reproducible activation change in the predicted direction; and selection by an outcome-blind stability rule rather than by favorable brain association. Our feature reports meet only some of these requirements, so the probe stands as a prospective protocol rather than evidence for any cortical feature.

P IMPLEMENTATION VERIFICATION AND REPRODUCIBILITY

Each result file logs its inputs, code hash, configuration fingerprint, model, story, participant, fold, feature bank, scaling statistics, selected penalty, held-out predictions, reliance profile, and null values; the cache rejects a file when any requested fingerprint differs. Verification occurs at three boundaries. Design tests check purging, participant/story separation, training-only scaling, and the training-mean R^2_{oos} denominator. Estimator tests compare ordinary and multiband ridge, held-out ablation, weighted row reduction, and both null generators with direct or synthetic calculations. Release tests check complete condition inventories, finite arrays, summary-to-array hashes, numerical identity across text, tables, and figures, and the absence of credentials, local paths, or author identity. Historical and exploratory artifacts are tagged by role, keeping them distinct from the artifacts behind the reported numbers.

The anonymous release contains the recorded analysis specification, path-neutral configuration, extraction and analysis code, direction- and fold-specific bank manifests, derived predictors permitted by model licenses, deidentified targets or reconstruction scripts, per-condition summaries and arrays, canonical JSON/CSV results, environment records, and SHA-256 manifests. Large source data, model weights, and disposable intermediates are not redistributed; the release lists the inputs needed to reconstruct them. Runtime is reported by scientific stage, with scheduler delay kept separate from algorithmic time; a staged rerun resumes only when configuration, code, input, and condition hashes match the recorded specification.